



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

**FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN MECHANICAL ENGINEERING**

EMM 100: ENGINEERING DRAWING AND DESIGN 1

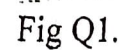
DATE: Friday 12th March 2021

TIME: 8.00 a.m. – 10.00 a.m.

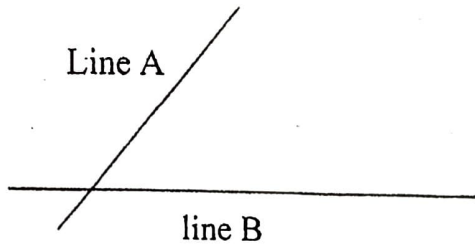
INSTRUCTIONS:

1. This paper contains five (5) questions
2. You are required to answer three (3) questions only
3. Question one (1) is compulsory and carries thirty (30) marks
4. The other four (4) questions carry twenty (20) marks each
5. Attempt any two (2) of the four questions
6. Construction lines should be faint and not be erased
7. All dimensions are in millimeters

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|-------|---|------------|
| (i) | A front elevation obtained by viewing the bracket in the direction of arrow F | (10 marks) |
| (ii) | An end view obtained by viewing the bracket in arrow E | (10 marks) |
| (iii) | A plan view projected from the front and end elevation. | (10 marks) |
| (iv) | A front elevation obtained by the section A-A | (10 marks) |



2. Given the location of the centre point with a major diameter of 60 cm and a minor diameter of 40 cm draw an ellipse using a concentric circle method. (10 marks)
3. Given an acute angle, lines A and B and a required radius of 20 cm ($r = 20$ cm), construct an arc tangent to an acute angle (less than 90°). (10 marks)



4. In reference to fig:Q4
- (i) Construct a four view drawing of this object (front, top, right, and left side views. (10 marks)
- (ii) Draw an oblique drawing of the item. (10 marks)

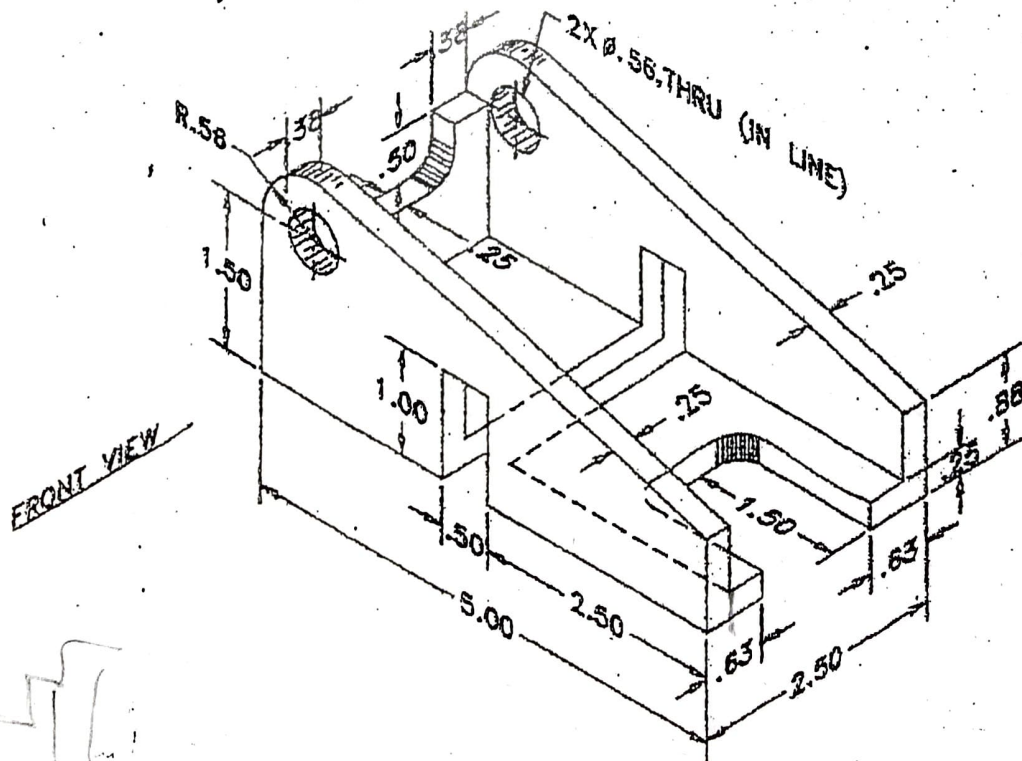


fig:Q4

5. Sketch an isometric view of a cube $50 \times 50 \times 50$ with a hole of $15 \text{ cm } \phi$ in the centre of one phase of the cube.

(10 marks)

- (i) Give an orthographic sketch of the cube.

(10 marks)

- (ii) Give an oblique sketch of the cube.